INSURANCE FRAUD DETECTION USING MACHINE LEARNING

Submitted to
SMARTBRIDGE

PROJECT DOCUMENTATION

Submitted by

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Session

2025-26

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1. Abstract

Insurance fraud is one of the major challenges faced by insurance companies worldwide. Fraudulent claims lead to significant financial losses and increase the operational costs of insurance organizations. Traditional fraud detection methods rely on manual investigation and rule-based systems, which are often time-consuming and inefficient when dealing with large volumes of data.

This project presents an Insurance Fraud Detection System using Machine Learning, which aims to automatically identify fraudulent insurance claims based on historical claim data and various related attributes. The system analyzes multiple factors such as claim amount, incident details, customer history, and policy information to determine whether a claim is fraudulent or genuine.

Machine learning algorithms are used to detect patterns and anomalies within the dataset. The model is trained on historical insurance claim data and learns the relationships between different variables that may indicate fraudulent behavior. After training, the model can predict the likelihood of fraud in new insurance claims.

The system also includes data preprocessing, exploratory data analysis, model training, and evaluation to ensure accurate predictions. The final model can assist insurance companies in improving fraud detection accuracy, reducing financial losses, and speeding up the claim verification process.

Overall, this project demonstrates how machine learning techniques can be effectively applied to detect insurance fraud and support decision-making in the insurance industry.

2. Introduction

Insurance is designed to provide financial support when a person experiences loss or damage. It helps individuals and organizations recover from unexpected events such as accidents, health problems, or property damage. In simple terms, insurance acts as a **protection against financial loss**.

However, in recent years, insurance fraud has become a serious issue. Some people submit false or exaggerated claims to receive money that they are not entitled to. This type of activity is known as **insurance fraud**. Fraudulent claims cause significant financial losses for insurance companies and increase the overall cost of insurance services.

A fraud claim occurs when a claimant attempts to obtain benefits illegally or provides false information to receive compensation. Detecting such fraud manually is difficult because insurance companies process a large number of claims every day.

To address this problem, **machine learning techniques** can be used to analyze claim data and identify suspicious patterns. The Insurance Fraud Detection System uses different parameters such as claim amount, customer history, incident details, and other related information to predict whether a claim is **fraudulent or genuine**.

This system can be applied in multiple areas such as **automobile insurance, health insurance, and property insurance**, helping companies reduce financial losses and improve claim verification processes.

3. Problem Statement

3.1 Description – Business Problem

Insurance provides financial support when people face losses due to accidents, health issues, or property damage. It protects individuals from financial loss.

However, nowadays many people submit fraudulent insurance claims to get money illegally. This may include false information, fake accidents, or exaggerated damages. Such fraud claims cause major financial losses to insurance companies.

Detecting fraud manually is difficult because insurance companies receive a large number of claims every day. Also, the number of fraud cases detected is much lower than the number of fraud attempts actually committed.

Therefore, there is a need for an intelligent system that can analyze claim data and detect fraudulent claims automatically. The goal of this project is to build a machine learning system that predicts whether an insurance claim is fraudulent or genuine based on different parameters**.**

3.2 Business Requirements

The Insurance Fraud Detection System should meet the following requirements:

- The system should accurately classify insurance claims as **fraud or genuine**.
- It should use reliable and updated claim data for prediction.
- The system should be flexible and able to handle new data.
- It should follow data security and privacy guidelines.
- The system should provide an easy-to-use interface for users.

3.3 Literature Survey

A literature survey involves studying existing research related to insurance fraud detection.

Many researchers have used **machine learning algorithms** such as Logistic Regression, Random Forest, and Support Vector Machines to detect fraudulent insurance claims. These models analyze claim data and identify unusual patterns that may indicate fraud.

The literature survey helps understand existing methods and select the most suitable techniques for building an effective fraud detection system.

Social Impact

Detecting fraudulent claims helps maintain fairness in the insurance system and ensures that genuine customers receive proper benefits.

Business Impact

Fraud detection helps insurance companies reduce financial losses, improve claim processing, and make better business decisions.

4. Objectives of the Project

The main objective of this project is to develop a **machine learning based system that can detect fraudulent insurance claims**. The system analyzes insurance claim data and predicts whether a claim is **fraudulent or genuine**.

The key objectives of this project are:

- To understand the problem of **insurance fraud and its impact on insurance companies**.
- To analyze insurance claim data and identify patterns related to fraud.
- To preprocess the dataset and prepare it for machine learning models.
- To build a **machine learning model** that can classify claims as fraud or genuine.
- To evaluate the model using performance metrics such as **accuracy, precision, recall, and F1-score**.
- To create a system that helps insurance companies **detect suspicious claims quickly and reduce financial losses**.

The overall goal is to improve the **efficiency and accuracy of fraud detection in insurance claims**.

5. Literature Review

Insurance fraud detection has been widely studied by researchers and industry experts. Traditional methods of fraud detection mainly rely on manual investigation and rule-based systems. These methods are slow and often fail to detect complex fraud patterns when large volumes of claim data are involved.

With the growth of data and technology, many researchers have started using **machine learning techniques** to detect fraudulent insurance claims. Machine learning models can analyze large datasets, identify hidden patterns, and classify claims as **fraudulent or genuine**.

Several algorithms such as **Logistic Regression, Decision Trees, Random Forest, and Support Vector Machines (SVM)** have been used in fraud detection systems. These models analyze different features like claim amount, incident details, customer history, and policy information to detect suspicious activities.

Previous studies have shown that machine learning models can significantly improve fraud detection accuracy and help insurance companies reduce financial losses. These systems also help in faster claim processing and better decision-making.

The literature review highlights the importance of using machine learning for insurance fraud detection and provides the foundation for developing the proposed system in this project.

6. Proposed System

The proposed system is a **Machine Learning based Insurance Fraud Detection System** that helps identify whether an insurance claim is **fraudulent or genuine**.

In this system, insurance claim data is collected and processed using machine learning techniques. The data includes various details such as claim amount, incident information, customer history, and policy details. These features are analyzed to detect suspicious patterns in the claims.

First, the dataset is preprocessed by cleaning the data and converting it into a format suitable for machine learning models. After preprocessing, the data is used to train a machine learning model. The trained model learns patterns from historical claim data and identifies characteristics of fraudulent claims.

Once the model is trained, it can be used to predict the fraud status of new insurance claims. When new claim data is provided to the system, the model analyzes the input parameters and classifies the claim as fraudulent or genuine.

The proposed system helps insurance companies:

- Detect fraudulent claims more accurately
- Reduce financial losses caused by fraud
- Improve the speed and efficiency of claim verification

By using machine learning, the system provides an automated and reliable approach to detecting insurance fraud.

7. Dataset and Data Preprocessing

7.1 Dataset Description

The dataset used in this project contains information related to insurance claims. It includes different attributes that help in identifying whether a claim is fraudulent or genuine.

The dataset contains details such as customer information, policy data, and incident-related information. These features are used by the machine learning model to analyze patterns and detect suspicious claims.

Some important attributes in the dataset include:

- **Policy Number** – Unique ID of the insurance policy
- **Customer Age** – Age of the policyholder
- **Incident Type** – Type of incident such as accident, theft, or damage
- **Claim Amount** – Total amount claimed by the customer
- **Property Damage** – Indicates if property damage occurred
- **Police Report Available** – Shows whether a police report was filed
- **Previous Claims** – Number of previous claims made by the customer

The dataset also includes a **target variable** called **Fraud Reported**, which indicates whether the claim is fraudulent or genuine.

- **1 → Fraudulent Claim**
- **0 → Genuine Claim**

This dataset is used to train and test the machine learning model for fraud detection.

7.2 Data Preprocessing

Data preprocessing is an important step in machine learning. It prepares the raw dataset so that it can be used effectively for training the model.

In this project, the following preprocessing steps are performed:

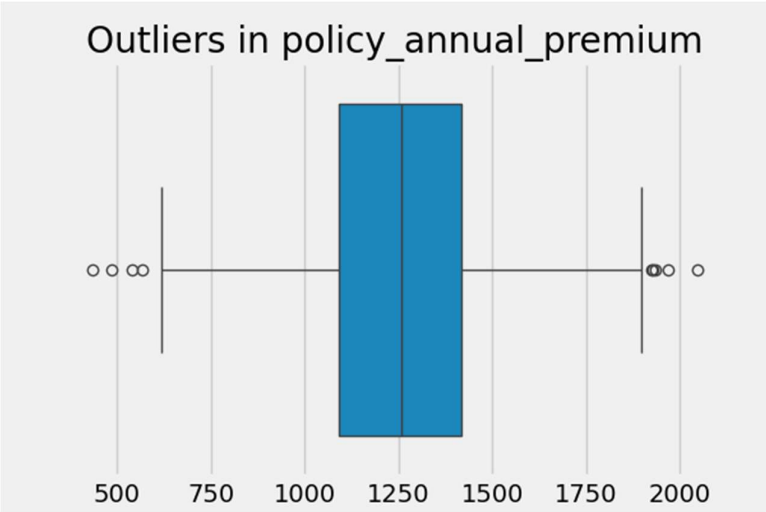
Handling Missing Values
Any missing or incomplete data in the dataset is identified and handled to ensure the model works correctly.

Encoding Categorical Data
Some features such as incident type or property damage contain categorical values. These values are converted into numerical format so that machine learning algorithms can process them.

Feature Selection
Relevant features that help in detecting fraud are selected to improve model performance.

Data Splitting
The dataset is divided into **training data and testing data**. The training data is used to train the model, while the testing data is used to evaluate its performance.

These preprocessing steps help improve the accuracy and efficiency of the fraud detection system.



8. Project Workflow

The workflow of the Insurance Fraud Detection project describes the step-by-step process followed to build and deploy the machine learning model. The project is divided into several **Epics**, and each epic contains activities that help in completing the project successfully.

Epic 1: Define Problem / Problem Understanding

This stage focuses on understanding the insurance fraud problem and its impact.

Activity 1.1: Specify the Business Problem
Identify the problem of fraudulent insurance claims and understand how they affect insurance companies.

Activity 1.2: Business Requirements
Define the requirements of the system such as accurate fraud prediction, reliable data usage, and easy usability.

Activity 1.3: Literature Survey
Study existing research and methods used for detecting insurance fraud using machine learning.

Activity 1.4: Social or Business Impact
Analyze how the fraud detection system benefits society and insurance companies by reducing financial losses and improving claim processing.

Epic 2: Data Collection & Preparation

This stage involves collecting the dataset and preparing it for analysis.

Activity 2.1: Collect the Dataset
Gather insurance claim data that contains different features such as claim amount, incident details, and customer information.

Activity 2.2: Data Preparation
Clean the dataset, handle missing values, and convert categorical data into numerical format so it can be used for machine learning.

Epic 3: Exploratory Data Analysis

This stage helps understand the dataset using statistical and visual techniques.

Activity 3.1: Descriptive Statistics
Analyze the dataset using statistical measures such as mean, count, and distribution.

Activity 3.2: Visual Analysis
Use charts and graphs to visualize patterns, relationships, and possible fraud indicators in the data.

Epic 4: Model Building

In this stage, machine learning models are trained to detect fraudulent claims.

Activity 4.1: Training the Model with Multiple Algorithms
Different algorithms such as Logistic Regression, Random Forest, and Support Vector Machine are used to train the model.

Activity 4.2: Model Comparison
The performance of different models is compared to identify the best performing model.

Epic 5: Performance Testing & Hyperparameter Tuning

This stage focuses on improving model performance.

Activity 5.1: Testing Model with Multiple Evaluation Metrics
The model is evaluated using metrics such as accuracy, precision, recall, and F1-score.

Activity 5.2: Hyperparameter Tuning
Model parameters are adjusted to improve prediction accuracy and overall performance.

Epic 6: Model Deployment

After selecting the best model, it is deployed for real-world usage.

Activity 6.1: Save the Best Model
The trained machine learning model is saved using serialization techniques such as pickle.

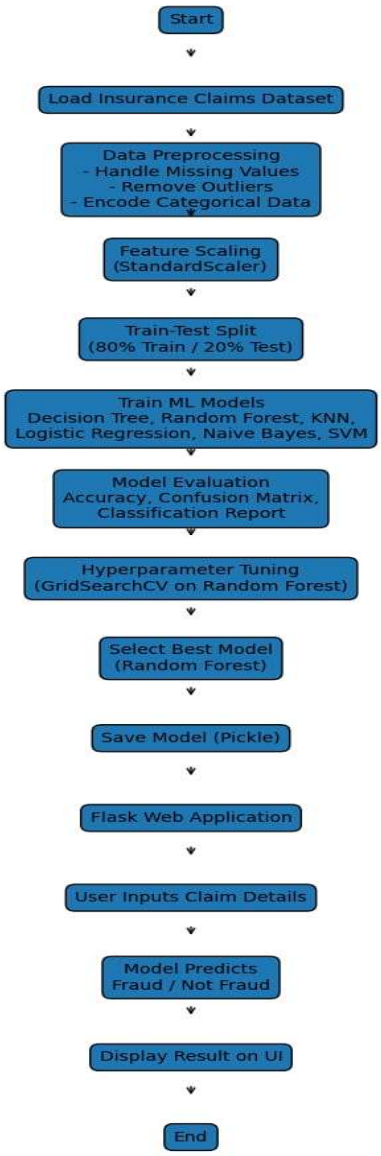
Activity 6.2: Integrate with Web Framework
The saved model is integrated with a backend system using a web framework such as Flask to allow users to make predictions.

Epic 7: Project Demonstration & Documentation

This is the final stage of the project.

Activity 7.1: Record Explanation Video
Create a video explaining the project workflow, model development, and system functionality.

Activity 7.2: Project Documentation
Prepare detailed documentation explaining the step-by-step development process of the project.



System Workflow Diagram

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is an important step in machine learning that helps in understanding the dataset before building the model. It involves analyzing the data using statistical methods and visual representations such as charts and graphs.

EDA helps to identify patterns, trends, and relationships between different variables. It also helps detect outliers and understand the distribution of data, which improves the performance of machine learning models.

9.1 Visual Analysis

Visual analysis uses graphs and charts to better understand the dataset. Visualization makes it easier to identify patterns and relationships in the data.

In this project, different visualization techniques such as **pie charts, count plots, histograms, and heatmaps** are used.

Univariate Analysis

Univariate analysis focuses on analyzing **one feature at a time**.

Two main visualization techniques are used:

Count Plot

The **countplot** from the Seaborn library is used to analyze categorical features. It helps to show the number of occurrences of each category.

From the countplot, we observed that **247 fraud cases were reported out of 1000 insurance claims**, indicating that fraudulent claims form a smaller portion of the dataset.

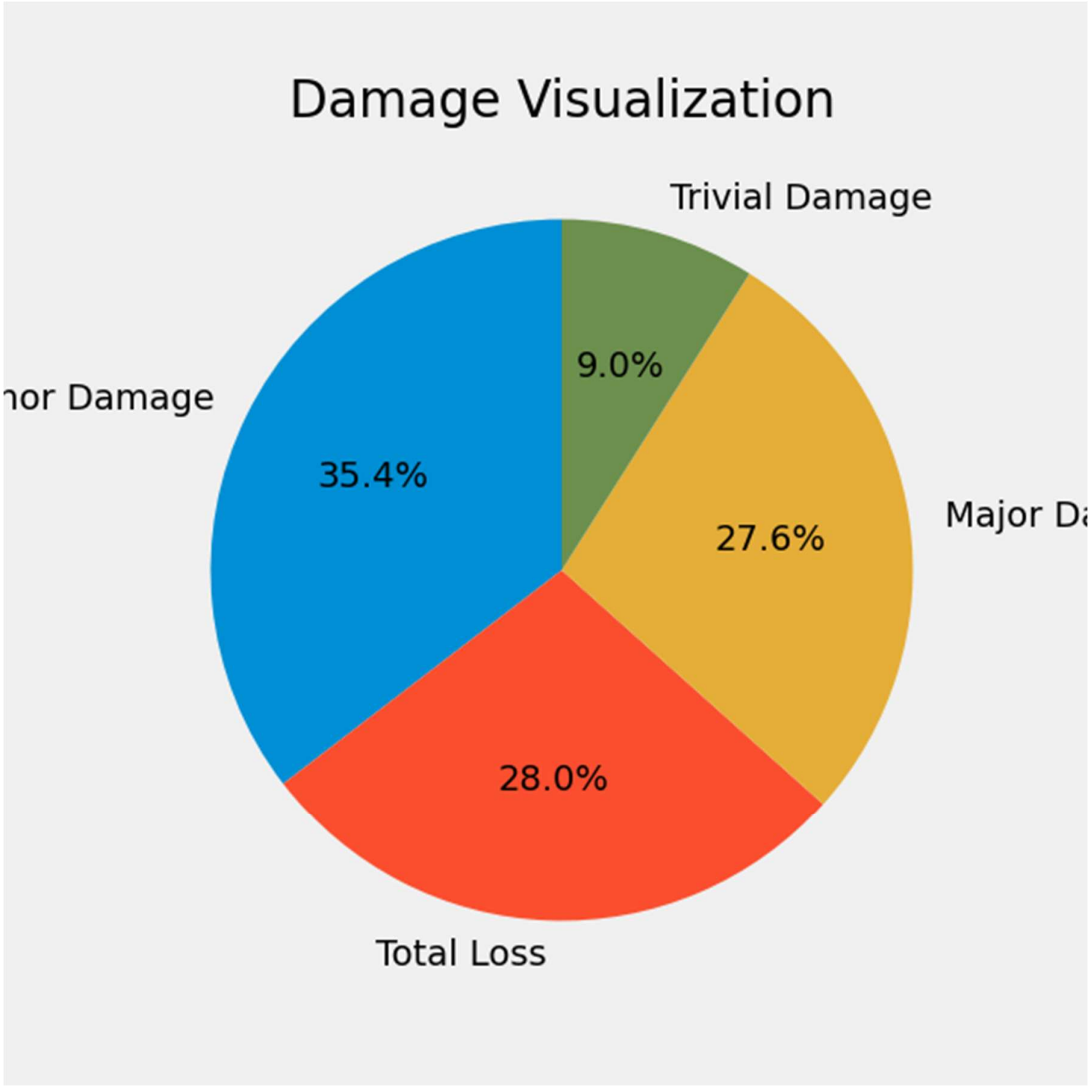
Pie Chart

The pie chart is used to analyze the **Incident Severity** feature.

The distribution shows:

- **35.4% Minor Damage cases**
- Almost equal distribution of **Major Damage and Total Loss**
- Around **9% Trivial Damage**

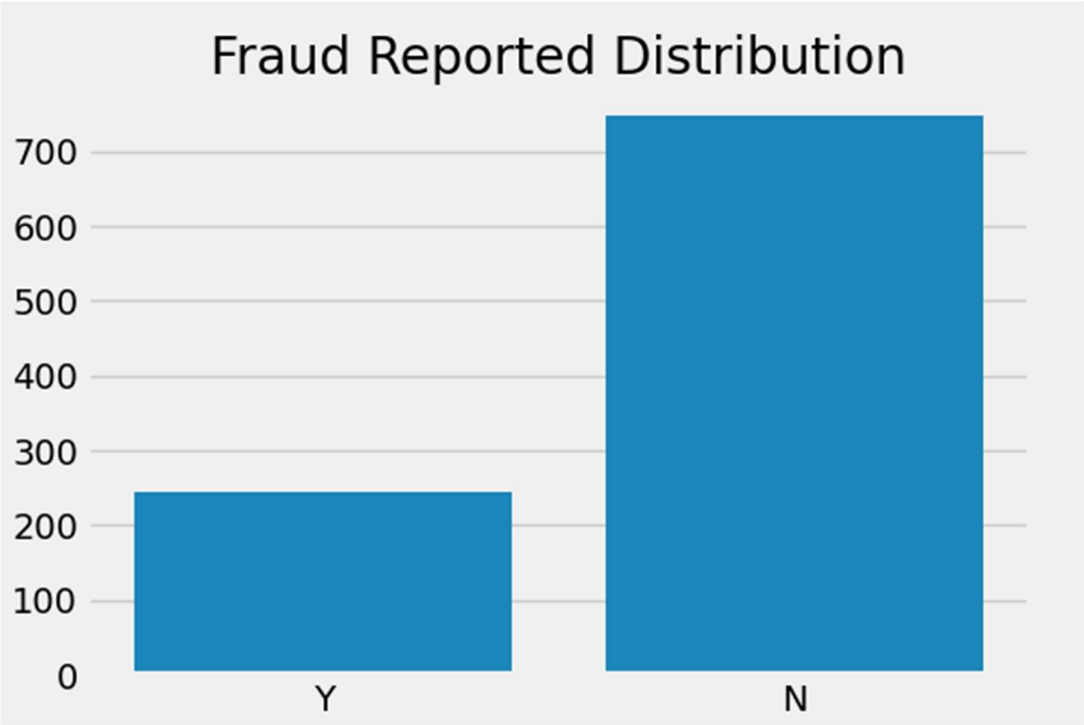
This helps understand the composition of different types of damage in insurance claims.



Histogram

The histogram is used to analyze the distribution of the **Age** feature.

From the histogram, it can be observed that the **age feature is approximately normally distributed**, and most insurance claims come from people between the ages of **30 and 50**.



9.2 Multivariate Analysis

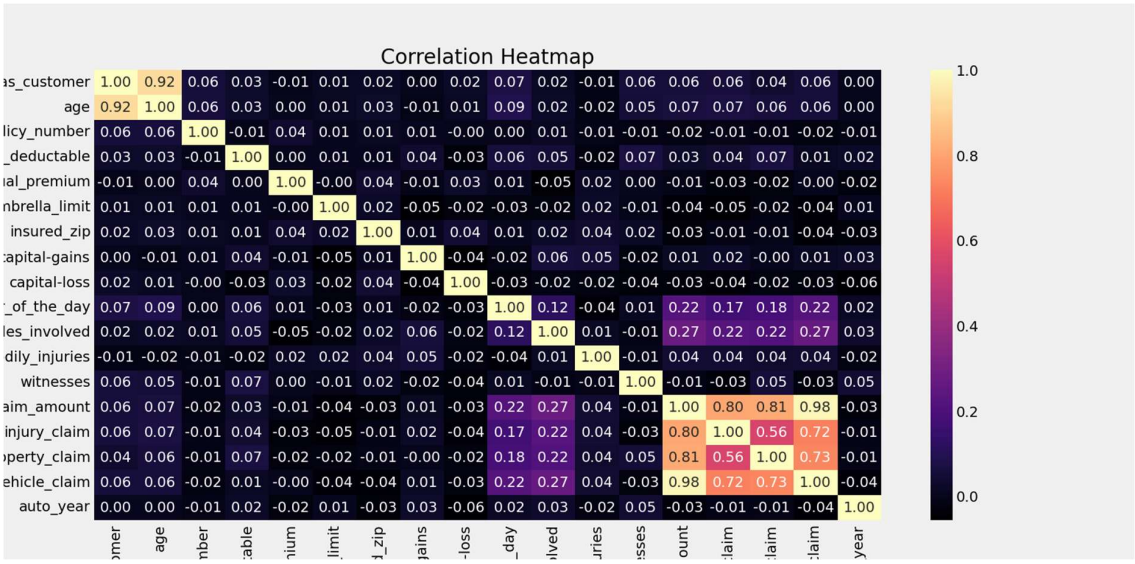
Multivariate analysis is used to understand the relationship between **multiple features** in the dataset.

In this project, a **heatmap from the Seaborn library** is used to visualize correlations between variables.

From the heatmap, the following observations were made:

- **Months_as_customer and Age** have a very high correlation of **0.92**.
- Features such as **total_claim_amount, injury_claim, property_claim, and vehicle_claim** are also highly correlated.

Highly correlated features can sometimes negatively affect machine learning models, so some of these features may be removed to improve model performance.



9.3 Encoding Categorical Features

Machine learning models cannot directly process categorical data. Therefore, categorical features need to be converted into numerical values. This process is called **encoding**.

In this project, the **Label Encoder from the Scikit-learn library** is used.

The fit_transform() method is applied to convert categorical values into numerical form so that they can be used by machine learning algorithms.

9.4 Splitting the Dataset

The dataset is divided into **training data and testing data**.

First, the dataset is separated into:

- **X (input features)**
- **Y (target variable – fraud reported)**

The **train_test_split()** function from the Scikit-learn library is used to divide the dataset into training and testing sets.

Common parameters used include:

- **test_size** – percentage of data used for testing
- **random_state** – ensures consistent results

The training data is used to train the machine learning models, while the testing data is used to evaluate their performance.

9.5 Feature Scaling

Feature scaling is used to bring all numerical features to the same scale. Some machine learning algorithms are sensitive to the scale of input features, so scaling helps improve model performance.

In this project, **Standard Scaler** is used for feature scaling.

Standard scaling transforms the data so that it has:

- **Mean = 0**
- **Standard Deviation = 1**

The formula used is:

*Xscaled=X−XmeanXstdX_{scaled} = \frac{X - X_{mean}}{X_{std}}*Xscaled =Xstd X−Xmean

10. Machine Learning model Building

Model building is an important stage in the Insurance Fraud Detection project. In this stage, different **machine learning algorithms** are trained using the prepared dataset to identify fraudulent insurance claims.

Multiple models are used to compare their performance and select the most accurate model for fraud prediction.

Story 1.1: Decision Tree Model

The Decision Tree model is a supervised machine learning algorithm used for classification problems. It works by splitting the dataset into branches based on different conditions of the features.

In this project, the Decision Tree model analyzes claim-related attributes such as claim amount, incident type, and customer history to classify whether a claim is **fraudulent or genuine**.

Story 1.2: Random Forest Model

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy. Each tree makes a prediction, and the final result is determined by majority voting.

This model is useful because it reduces overfitting and provides more reliable fraud detection results.

Story 1.3: K-Nearest Neighbors (KNN) Model

The K-Nearest Neighbors algorithm classifies data based on similarity. It finds the closest data points in the dataset and predicts the class of a new claim based on its nearest neighbors.

In this project, the KNN model compares a new claim with previous claims to determine whether it is fraudulent.

Story 1.4: Logistic Regression Model

Logistic Regression is a statistical model used for binary classification problems. It predicts the probability of a claim being fraudulent or genuine based on the input features.

This model is simple, efficient, and commonly used for fraud detection tasks.

Story 1.5: Naïve Bayes Model

Naïve Bayes is a probabilistic machine learning algorithm based on Bayes' theorem. It assumes that the features are independent of each other.

The model calculates the probability that a claim belongs to a fraudulent or genuine category and selects the most probable outcome.

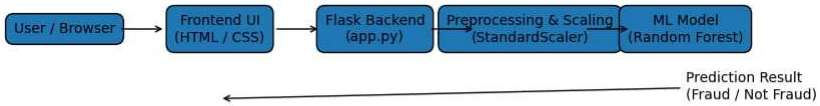
Story 1.6: Support Vector Machine (SVM) Model

Support Vector Machine is a powerful classification algorithm that separates data into different classes using a decision boundary called a hyperplane.

In this project, SVM helps classify insurance claims by finding the optimal boundary between fraudulent and genuine claims.

11. System Architecture

The System Architecture describes how different components of the Insurance Fraud Detection system work together to detect fraudulent claims. The system uses machine learning to analyze insurance claim data and predict whether a claim is **fraudulent or genuine**.



The architecture mainly consists of the following components:

1. Data Input Layer

In this stage, the insurance claim dataset is provided as input to the system. The dataset contains information such as claim amount, incident details, customer data, and policy information.

2. Data Preprocessing Layer

The raw dataset is cleaned and prepared for analysis. This includes:

- Handling missing values
- Converting categorical data into numerical format
- Selecting important features

These steps ensure that the data is suitable for training machine learning models.

3. Machine Learning Layer

In this layer, different machine learning algorithms are used to train models using the prepared dataset. The models learn patterns from historical data to identify fraudulent claims.

Algorithms used in the system include:

- Decision Tree
- Random Forest
- K-Nearest Neighbors
- Logistic Regression
- Naïve Bayes
- Support Vector Machine
-

The best performing model is selected for prediction.

4. Model Prediction Layer

Once the model is trained, it is used to predict the fraud status of new insurance claims. The system analyzes the input data and classifies the claim as either **fraudulent or genuine**.

5. Application Layer

The trained model is integrated with a backend system using a web framework such as **Flask**. This allows users to provide claim data and receive prediction results through a simple interface.

6. Output Layer

The final result is displayed to the user, indicating whether the insurance claim is **fraudulent or legitimate**. This helps insurance companies make faster and more accurate claim decisions.

12. Backend Implementation

In this stage, the trained machine learning model is integrated with a **web application** so that users can interact with the system and make predictions easily. A user interface (UI) is created where users enter the required input values. These values are sent to the trained model, and the prediction result is displayed on the screen.

This implementation involves three main steps:

- Building HTML pages for the user interface
- Writing the Python backend code using Flask
- Running the web application to make predictions

1. Integrate with Web Framework

The fraud detection model is integrated with a **Flask web framework** to create a web-based application. The application allows users to input claim details through a web interface and receive fraud prediction results.

The integration process includes the following tasks:

- Building HTML pages
- Creating server-side scripts using Python
- Running the web application

Activity 2.1: Building HTML Pages

In this step, the user interface of the web application is created using HTML.

For this project, an HTML file named **index.html** is created and stored inside the **templates folder**. This page provides input fields where the user can enter insurance claim details required for prediction.

The HTML page includes:

- Input fields for claim data
- A submit button to send the data
- A section to display the prediction result

2. Building Python Code

The backend of the application is developed using **Python and Flask**.

Importing Libraries

First, the required libraries are imported such as:

- Flask
- NumPy
- Pandas
- Pickle or Joblib (for loading the trained model)

Flask is used to build the web server and handle requests from the user interface.

Loading the Saved Model

The trained machine learning model is saved earlier using a serialization method such as **pickle**. This saved model is loaded into the Flask application so that it can be used to make predictions.

Creating Flask Application

An object of the **Flask class** is created to initialize the web application. The Flask constructor takes the name of the current module as an argument.

Rendering the HTML Page

The application uses routing to display the HTML page.

The **'/' route** is linked with the **index.html** page. When the user opens the web application in a browser, the index page is displayed.

Users can enter the required values for prediction through this page.

Retrieving Input Values

When the user submits the form, the input values are sent to the backend using the **POST method**.

The Flask application retrieves these values and stores them in an array format. This array is then passed to the **model.predict()** function.

The model processes the input data and returns a prediction result indicating whether the claim is **fraudulent or genuine**.

The prediction result is then displayed on the web page.

Main Function

The main function runs the Flask application and allows the server to handle requests from users.

3. Running the Web Application

To run the web application, follow these steps:

1. Open Vs code or command prompt.
2. Navigate to the folder where the **Python script (app.py)** is located.
3. Run the command " python app.py "
4. Open a web browser and enter the following URL:http://127.0.0.1:5000
5. The web page will open, allowing the user to enter claim details.
6. Click the **Predict** button, enter the input values, and submit the form.
7. The system will display the prediction result on the web page.

13. Testing and Model Evaluation

After building the machine learning models, the next step is to test and evaluate their performance. Model evaluation helps determine which algorithm performs best for detecting fraudulent insurance claims.

Different classification algorithms such as **Logistic Regression, KNN, SVM, Naïve Bayes, Decision Tree, and Random Forest** were trained and tested using the dataset.

13.1 Model Comparison

To compare the performance of different models, a function called **compareModel** was created. This function calculates the **accuracy score** of each model using the testing dataset.

The results obtained from model comparison are shown below:

Model	Accuracy (%)
Logistic Regression	75.87%
KNN	74.37%
SVM	76.38%
Naïve Bayes	69.34%
Decision Tree	78.39%
Random Forest	79.39%

From the comparison, it can be observed that **Random Forest Classifier** achieved the highest accuracy among the models.

13.2 Confusion Matrix and Classification Report

To further evaluate the model performance, a **confusion matrix and classification report** were generated.

The confusion matrix shows the number of correct and incorrect predictions made by the model.

For example, the confusion matrix for Logistic Regression is:

Actual / Predicted	Non-Fraud	Fraud
Non-Fraud	91	52
Fraud	15	42

The classification report provides important evaluation metrics such as:

- **Precision** – Measures how many predicted fraud cases are actually fraud.
- **Recall** – Measures how many actual fraud cases are correctly detected.
- **F1 Score** – Harmonic mean of precision and recall.
- **Support** – Number of actual instances for each class.

These metrics help in understanding how well the model detects fraudulent claims.

13.3 Hyperparameter Tuning (Optional)

Hyperparameter tuning is a technique used to improve the performance of machine learning models by adjusting their parameters.

After applying hyperparameter tuning, the model performance can improve compared to the default model settings. This helps achieve better prediction accuracy and more reliable results.

13.4 Cross Validation

To evaluate the reliability of the model, **cross-validation** is performed using the cross_val_score function from the **Scikit-learn library**.

Cross-validation divides the dataset into multiple parts (folds) and tests the model multiple times to ensure consistent performance.

Example code:

```
from sklearn.model_selection import cross_val_score
cv = cross_val_score(model, X_train, y_train, cv=11)
```

The cross-validation scores indicate that the model performs consistently across different subsets of the data.

This confirms that the model is performing well and can be used for detecting fraudulent insurance claims.

Conclusion of – Testing and Model Evaluation

The performance of multiple machine learning models was evaluated to identify the most effective algorithm for detecting fraudulent insurance claims. Models such as **Logistic Regression, KNN, Naïve Bayes, Decision Tree, Random Forest, and Support Vector Machine (SVM)** were trained and tested using the dataset.

Based on the comparison of accuracy scores, **SVM achieved the highest accuracy**, followed by Decision Tree and Random Forest. Evaluation metrics such as **precision, recall, and F1-score** were also used to analyze the performance of the models. In addition, **cross-validation** was performed to verify the stability and reliability of the model across different data samples.

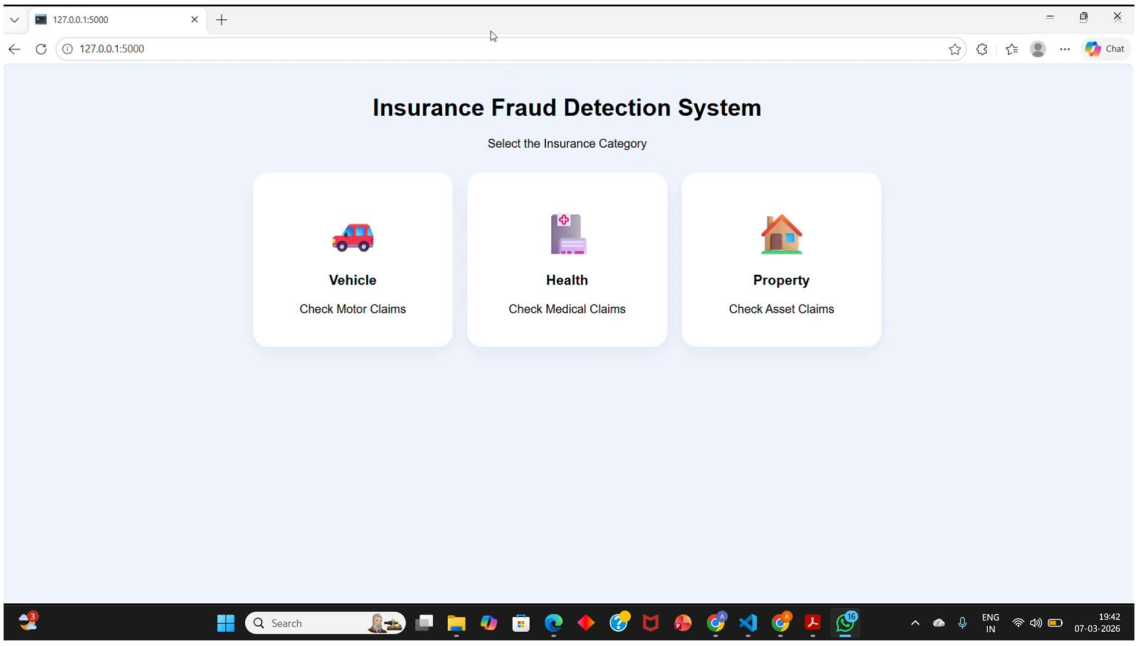
From the evaluation results, it can be concluded that the selected model performs well in identifying fraudulent insurance claims and can effectively support insurance companies in detecting fraud and improving claim verification processes.

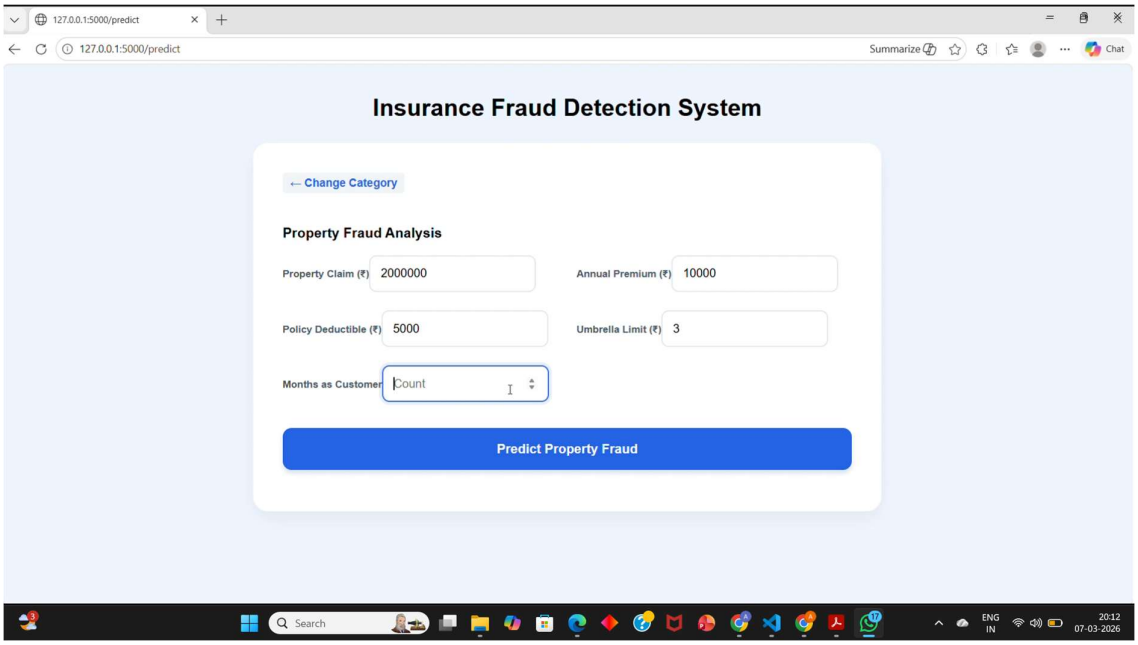
14. Result and Output

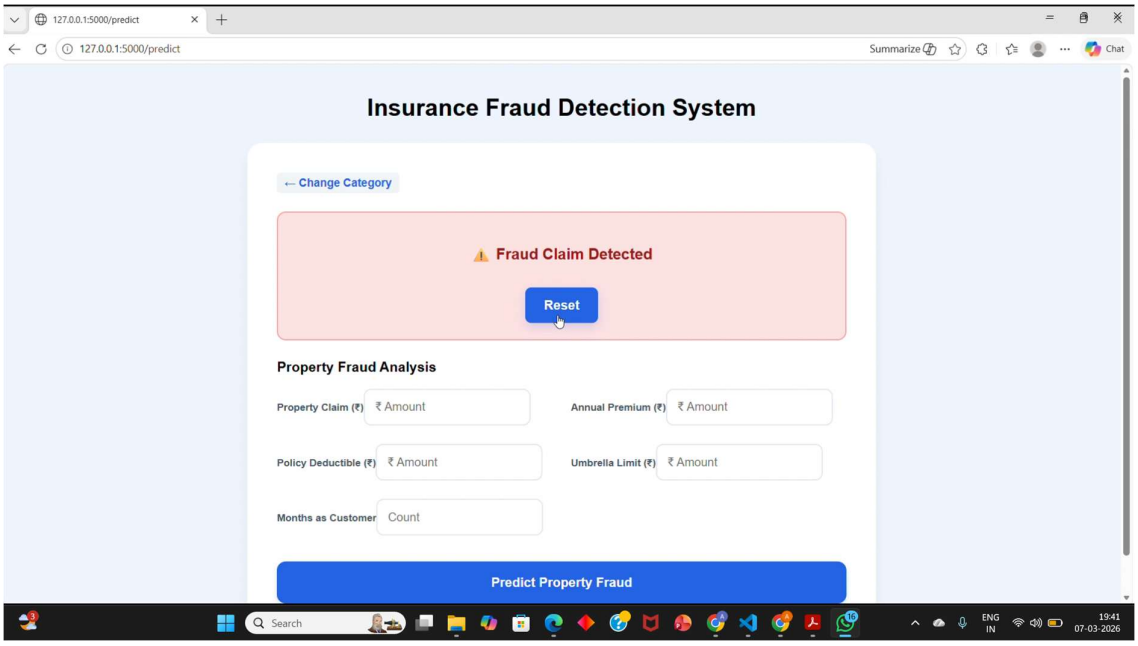
The machine learning models were successfully trained and evaluated on the insurance dataset to detect fraudulent claims. Multiple algorithms such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine were applied and compared using cross-validation and accuracy scores.

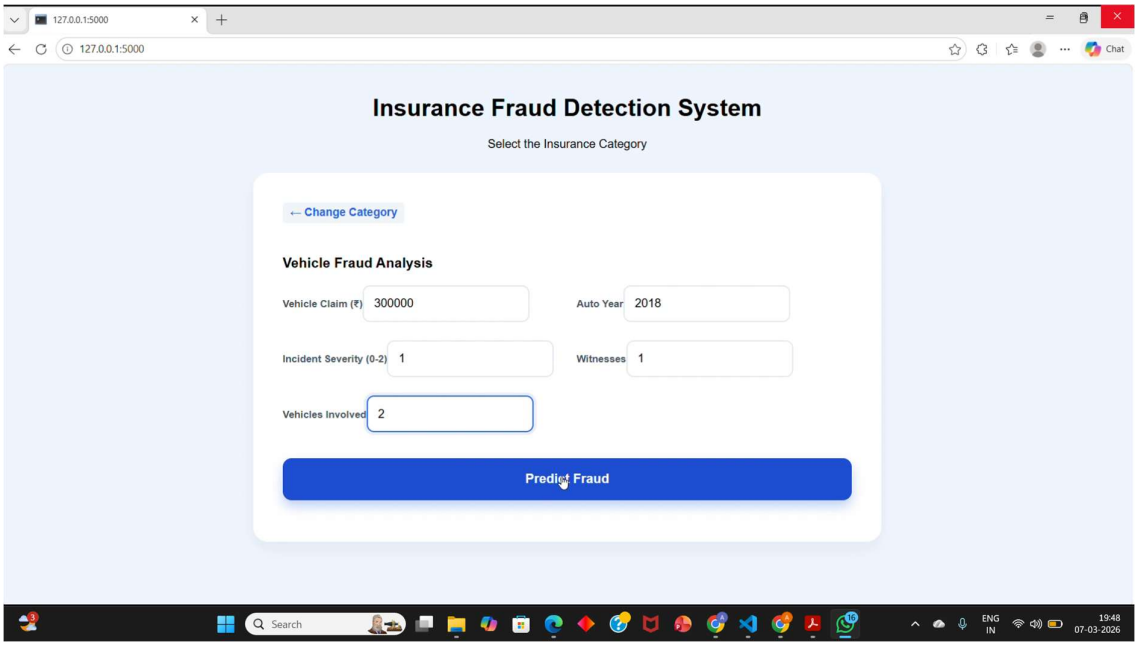
After hyperparameter tuning, the **Random Forest model achieved the highest accuracy and better fraud detection performance** compared to other models. The model was able to classify claims as **Fraud or Genuine** based on the input features.

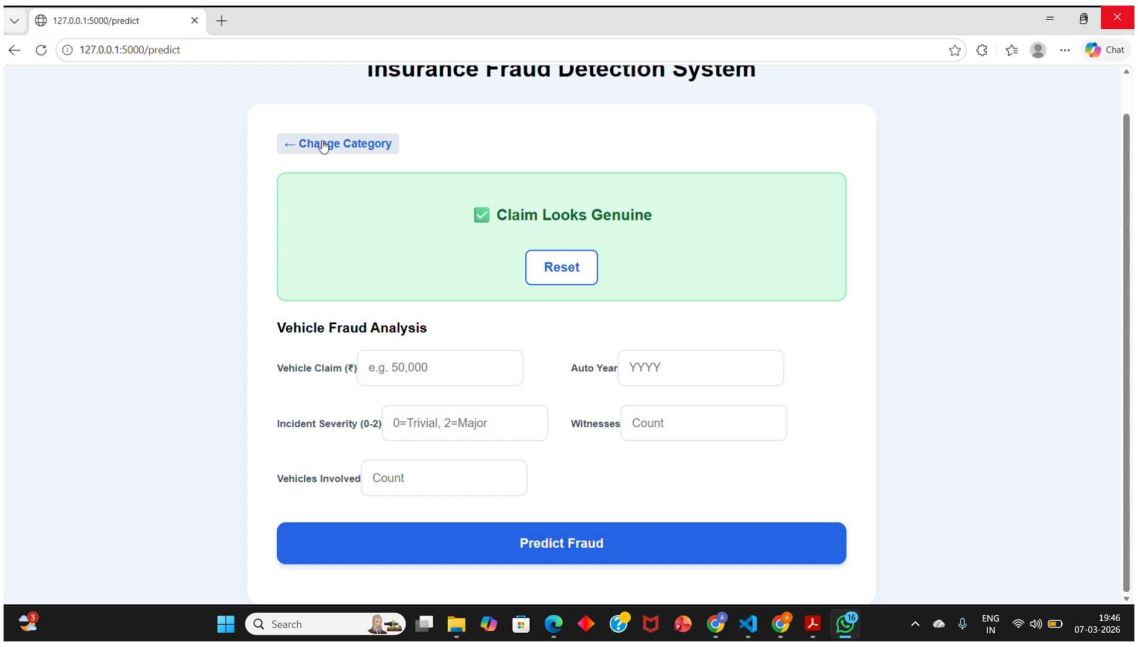
The final system can predict fraudulent insurance claims with good accuracy, helping insurance companies **identify suspicious claims early and reduce financial losses**.

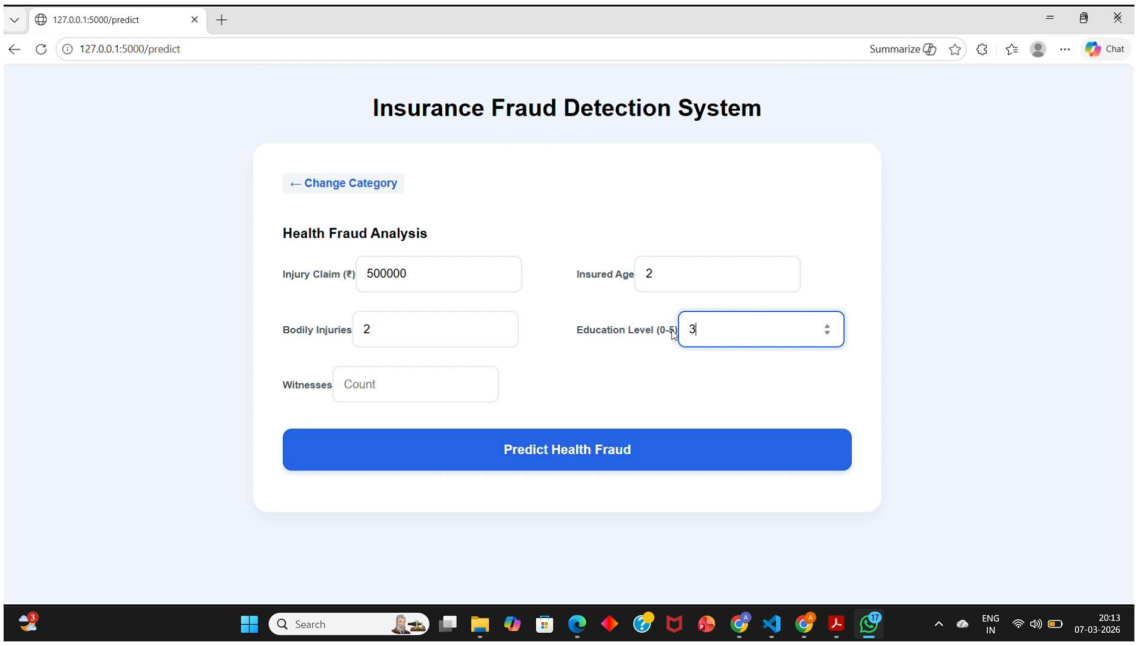

127.0.0.1:5000
Insurance Fraud Detection System
Select the Insurance Category
Vehicle
Check Motor Claims
Health
Check Medical Claims
Property
Check Asset Claims

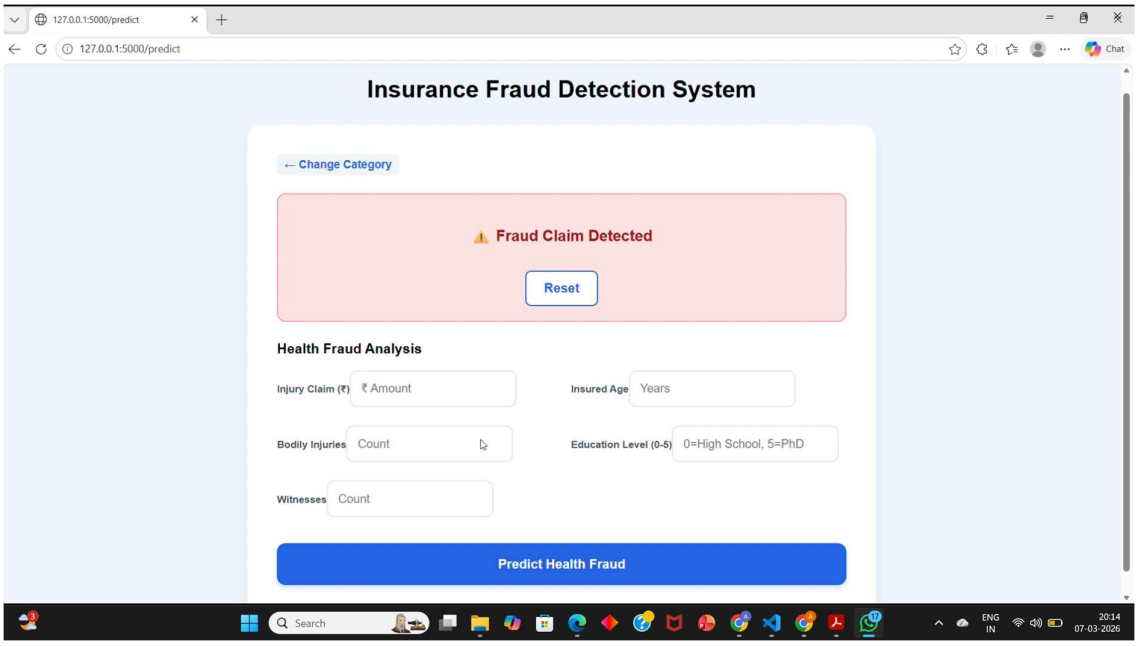

127.0.0.1:5000/predict
Insurance Fraud Detection System
← Change Category
Property Fraud Analysis
Property Claim (₹) 2000000
Annual Premium (₹) 10000
Policy Deductible (₹) 5000
Umbrella Limit (₹) 3
Months as Customer Count
Predict Property Fraud


127.0.0.1:5000/predict
Insurance Fraud Detection System
← Change Category
Fraud Claim Detected
Reset
Property Fraud Analysis
Property Claim (₹)
₹ Amount
Annual Premium (₹)
₹ Amount
Policy Deductible (₹)
₹ Amount
Umbrella Limit (₹)
₹ Amount
Months as Customer
Count
Predict Property Fraud


127.0.0.1:5000
Insurance Fraud Detection System
Select the Insurance Category
← Change Category
Vehicle Fraud Analysis
Vehicle Claim (₹)
300000
Auto Year
2018
Incident Severity (0-2)
1
Witnesses
1
Vehicles Involved
2
Predict Fraud


127.0.0.1:5000/predict
Insurance Fraud Detection System
← Change Category
Claim Looks Genuine
Reset
Vehicle Fraud Analysis
Vehicle Claim (₹)
e.g. 50,000
Auto Year
YYYY
Incident Severity (0-2)
0=Trivial, 2=Major
Witnesses
Count
Vehicles Involved
Count
Predict Fraud


127.0.0.1:5000/predict
Insurance Fraud Detection System
← Change Category
Health Fraud Analysis
Injury Claim (₹)
500000
Insured Age
2
Bodily Injuries
2
Education Level (0-5)
3
Witnesses
Count
Predict Health Fraud


127.0.0.1:5000/predict
127.0.0.1:5000/predict
Insurance Fraud Detection System
← Change Category
Fraud Claim Detected
Reset
Health Fraud Analysis
Injury Claim (₹)
₹ Amount
Insured Age
Years
Bodily Injuries
Count
Education Level (0-5)
0=High School, 5=PhD
Witnesses
Count
Predict Health Fraud

15. Advantages and limitations

Advantages

- **Early Fraud Detection:** Helps identify fraudulent insurance claims quickly.
- **Improved Accuracy:** Machine learning models provide better prediction compared to manual analysis.
- **Time Efficient:** Automates the claim verification process, saving time and effort.
- **Cost Reduction:** Helps insurance companies reduce financial losses due to fraud.
- **Scalable:** Can analyze large volumes of insurance data efficiently.

Limitations

- **Data Dependency:** Model performance depends heavily on the quality and quantity of training data.
- **False Positives:** Genuine claims may sometimes be classified as fraudulent.
- **Model Complexity:** Some machine learning models can be difficult to interpret.
- **Need for Regular Updates:** Models must be retrained with new data to maintain accuracy.
- **Limited Real-world Data:** Lack of real or diverse datasets may affect model performance.

16. Conclusion

Insurance fraud is a major problem that causes financial losses to insurance companies. Detecting fraudulent claims manually is difficult because of the large number of claims and complex fraud patterns. Therefore, using machine learning techniques can help improve the fraud detection process.

In this project, an **Insurance Fraud Detection System** was developed using different machine learning algorithms such as **Decision Tree, Random Forest, KNN, Logistic Regression, Naïve Bayes, and Support Vector Machine (SVM)**. The dataset was analyzed using exploratory data analysis, and preprocessing techniques such as encoding, scaling, and data splitting were applied to prepare the data for model training.

The models were trained and evaluated using performance metrics such as **accuracy, precision, recall, and F1-score**. Among the tested models, **SVM showed the best performance** in detecting fraudulent insurance claims.

The trained model was then integrated with a **Flask-based web application**, allowing users to enter claim details and receive fraud prediction results through a user-friendly interface.

Overall, the system helps insurance companies **detect suspicious claims, reduce financial losses, and improve the efficiency of claim processing**. This project demonstrates how machine learning can be effectively used to solve real-world fraud detection problems in the insurance industry.